

Statistical Decision Making 45-752

Broadening the Domain of Application of Regression Models

This file is named: 08_slides.pptx
Commands are in: 08_notebook.ipynb

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Stepwise Regression and Two Applications

Stepwise Regression can aid in selecting variables.

We will consider two quite different applications:

Hospital Satisfaction

Natural Gas Demand

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Application: Hospital Satisfaction

A group of Tepper students volunteered to serve as consultants to study factors that affect patients' satisfaction with their hospital stays.

The focus of the study was on the hospital environment, food, and related services (not on quality of medical care).

The students conducted a survey of a sample of 56 patients.

Patients were asked to rate a variety of aspects of the hospital environment and services.

These are listed on the next slide.

The goal of the study was to identify factors that were of greatest importance in influencing patient satisfaction.

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Hospital Satisfaction Data

Survey of 56 patients:

OVERALL: Overall satisfaction with inpatient experience (5-point scale)

Information about patient:

- GENDER
- Age

Variables rated on 5-point scale:

- ACCOMODATIONS: Overall satisfaction with accommodations
- IMPACCOMO: Importance of accommodations
- RoomBEDDING
- RoomCLEANLINESS
- RoomDECOR
- RoomFAMILY
- RoomTEMPERATURE:
- RoomTV
- RoomNoise_In: Noise in room
- RoomNoise_out: Noise Outside room

- RoomPRIVACY
- RECENT
- RoomRESTROOM
- RoomStaffcourtesy
- MEALSatisfact: Overall satisfaction with meals.
- IMPMEALS: Importance of meals.
- MealCOURTESY
- MealTASTE
- MealTEMPERATURE
- MealTIMING
- MealVARIETY
- WOW

Indicators for private room, bathroom:

- ROOM: Private=1, Shared=0
- BATHROOM: Private=1, Shared=0

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View Some Histograms



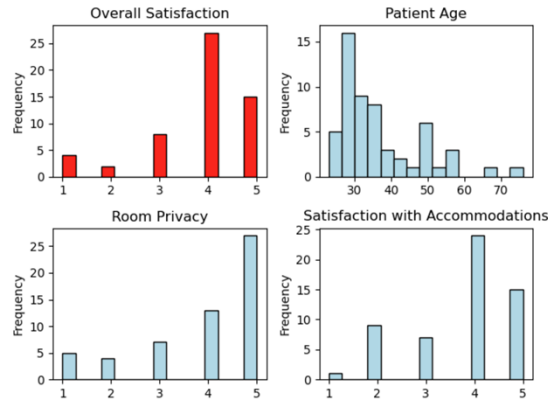
The dependent variable is Overall Satisfaction

Histograms of a few right hand-side variables are also shown here.

We see that there is a wide range of patient ages with relatively few older patients.

The bottom two histograms show the ratings of satisfaction with room privacy and satisfaction with accommodations.

For practice, you might want to view histograms of some other variables.



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View Summary Statistics



I issued the command:

```
# View summary statistics transposed
Hospital_Satisfaction.describe().transpose()
```

The above command transposes the output to the format shown at the right.

-We see mean Overall satisfaction is 3.84 with 50% rating Overall at or above 4.0.

-RoomCleanliness has the highest mean rating.

-RoomFamily has the lowest mean rating.

Note: MealFactors_Imp is about importance of meals, not a satisfaction measure.

	count	mean	std	min	25%	50%	75%	max
OVERALL	56.0	3.839286	1.091734	1.0	3.75	4.0	5.00	5.0
ACCOMMODATIONS	56.0	3.767857	1.078569	1.0	3.00	4.0	5.00	5.0
AGE	56.0	36.696429	11.534801	23.0	29.00	33.0	42.25	76.0
GENDER	56.0	0.642857	0.483494	0.0	0.00	1.0	1.00	1.0
IMPACCOMO	56.0	4.071429	0.531414	1.0	4.00	4.0	5.00	5.0
IMPMEALS	56.0	3.678571	0.833550	1.0	3.00	4.0	4.00	5.0
MEALCOURTESY	56.0	3.785714	1.056827	1.0	3.00	4.0	5.00	5.0
MEALFACTORS_IMP	56.0	2.267857	0.797521	1.0	2.00	2.0	3.00	5.0
MEALSATISFACT	56.0	3.232143	1.175370	1.0	2.00	3.0	4.00	5.0
MEALTASTE	56.0	3.232143	1.078569	1.0	2.00	3.0	4.00	5.0
MEALTEMP	56.0	3.571429	1.203890	1.0	3.00	4.0	4.25	5.0
MEALTIMING	56.0	3.375000	1.121079	1.0	3.00	3.0	4.00	5.0
MEALVARIETY	56.0	3.142857	1.118905	1.0	2.00	3.0	4.00	5.0
RECENT	56.0	4.607143	1.122960	1.0	4.00	5.0	5.00	6.0
ROOMBEDDING	56.0	3.392857	1.185957	1.0	2.75	4.0	4.00	5.0
ROOMCLEANLINESS	56.0	4.071429	1.076309	1.0	3.75	4.0	5.00	5.0
ROOMDECOR	56.0	3.357143	1.285281	1.0	3.00	3.0	4.00	5.0
ROOMFAMILY	56.0	2.642857	1.380758	1.0	1.00	2.0	4.00	5.0
ROOMNOISE_IN	56.0	3.910714	1.225142	1.0	3.00	4.0	5.00	5.0
ROOMNOISE_OUT	56.0	2.984286	1.393748	1.0	2.00	3.0	4.00	5.0
ROOMPRIVACY	56.0	3.946429	1.313145	1.0	3.00	4.0	5.00	5.0
ROOMRESTROOM	56.0	3.642857	1.118905	1.0	3.00	4.0	4.25	5.0
ROOMSTAFFCOURTESY	56.0	4.035714	0.952781	1.0	4.00	4.0	5.00	5.0
ROOMTEMPERATURE	56.0	3.500000	1.206045	1.0	3.00	4.0	4.00	5.0
ROOMTV	56.0	3.250000	1.209808	1.0	2.00	3.0	4.00	5.0
WOW	56.0	1.785714	0.779610	1.0	1.00	2.0	2.00	4.0

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Regression with All Variables.



The dependent variable is: Overall Satisfaction with hospital stay.

There are 56 observations and 22 variables.

Are any right-hand-side variables significant?

- At the 5% level?
- At the 10% level?

These p-values look discouraging.

But..... A Wald test of the hypothesis that coefficients of all variables except the intercept are equal to zero yields a **p-value=.0255**.

The regression output always includes a Wald test of the null hypothesis that all population coefficients equal 0 except the intercept. Please remember this!

The low p-value for the Wald test provides some reason for encouragement.

Dep. Variable:	OVERALL	R-squared:	0.470				
Model:	OLS	Adj. R-squared:	0.116				
No. Observations:	56	F-statistic:	2.109				
Covariance Type:	h0	Prob (F-statistic):	0.0255				
	Intercept	1.5895	1.719	0.882	0.384	-1.070	-4.989
ACCOMMODATIONS	0.3128	0.181	1.793	0.062	-0.054	0.682	0.037
AGE	2.799e-05	0.018	0.002	0.999	-0.037	0.037	0.037
GENDER	0.2276	0.389	0.086	0.942	-0.363	0.363	1.018
MEALCOURTESY	0.2241	0.201	1.116	0.272	-0.184	0.184	0.633
MEALSATISFACT	-0.2791	0.272	-1.028	0.312	-0.832	0.832	0.273
MEALTASTE	0.1588	0.238	0.668	0.508	-0.324	0.324	0.642
MEALTEMP	0.0812	0.210	-0.387	0.705	-0.312	0.312	0.144
MEALTIMING	-0.0148	0.142	-0.104	0.918	-0.304	0.304	0.276
MEALVARIETY	-0.0283	0.166	-0.123	0.903	-0.357	0.357	0.317
RECENT	-0.0535	0.145	-0.368	0.715	-0.349	0.349	0.242
ROOMBEDDING	0.2918	0.281	1.037	0.307	-0.280	0.280	0.864
ROOMCLEANLINESS	-0.1838	0.198	-0.830	0.359	-0.586	0.586	0.218
ROOMDECOR	-0.1036	0.182	-0.566	0.575	-0.476	0.476	0.269
ROOMFAMILY	-0.1116	0.150	-0.887	0.341	-0.267	0.267	0.146
ROOMNOISE_IN	-0.0429	0.289	-0.165	0.870	-0.312	0.312	0.487
ROOMNOISE_OUT	0.0358	0.121	0.296	0.769	-0.210	0.210	0.282
ROOMPRIVACY	0.1300	0.287	0.573	0.371	-0.391	0.391	0.697
ROOMRESTROOM	0.1364	0.265	1.195	0.240	-0.222	0.222	0.855
ROOMSTAFFCOURTESY	0.0465	0.200	0.332	0.742	-0.349	0.349	0.473
ROOMTEMPERATURE	-0.0806	0.126	-0.682	0.500	-0.341	0.341	0.170
ROOMTV	-0.1235	0.144	-0.360	0.396	-0.416	0.416	0.169
WOW	-0.1054	0.279	-0.499	0.621	-0.708	0.708	0.429

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What do we make of previous regression?



We have 56 observations. We have 22 variables, and the intercept.

Estimating 23 coefficients with only 56 observations is asking a lot from a limited amount of data.

Notice the contrast of the R^2 and the R^2_{adj} in the output on the previous slide. This difference between R^2 and R^2_{adj} arises because the right-hand-side variables together explain substantial variance in the dependent variable, but few if any right-hand-side variables are significant.

The Wald test that all coefficients are zero except the intercept yields a p-value of .0255. This provides some encouragement that the variables collectively have predictive value.

Ideally, we would like to have much more data.

Can we draw any tentative conclusions from the information we have?

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Choosing Variables



In some applications, you will have no difficulty deciding which variables to include in your model, e.g., diamond data.

In other applications, you have many variables that you can potentially include. You face a decision about which ones to select.

- There may be some variables that you will definitely wish to keep in your model.
- Other variables, you may be less sure about keeping.

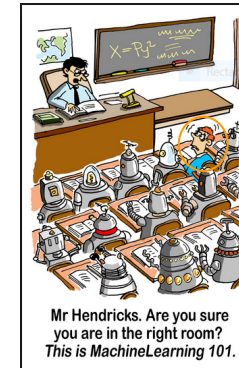
There are methods that can help you.

One such method is stepwise regression.

The appendix provides a brief description of [other methods](#).

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Stepwise Regression



There are several variants of stepwise regression. All have in common that they use a criterion of model fit for selecting variables.

Here is one approach. Let SSE denote the sum of squared errors in a regression.

Suppose we have 10 candidate independent variables.

We run 10 regressions, trying each variable one at a time. We keep the variable that gives the smallest p-value.

We then run 9 regressions, keeping the first variable we selected and including each of the remaining 9 variables one at a time. As our second variable, we choose the one that, paired with our first variable, gives the smallest SSE. We now have selected two variables.

We then run 8 regressions, keeping the first two variables in all 8 regressions while including each remaining variable one at a time. We keep as our third variable the one that, combined with the first two variables, gives the smallest SSE. We now have three variables.

Keep going until there is no remaining variable that reduces the SSE by more than a specified amount. This is called a “stopping rule”.

The method we use applies a similar approach, as detailed on the next slide.

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Stepwise Regression



We will use the stepwise method that selects variables based on p-values of coefficients.

Regressions are estimated with OVERALL as the dependent variable.

Each independent variable is included individually. Hence, 22 regressions are run, each with a different explanatory variable.

The variable with the lowest p-value from these 22 regressions is retained.

In our application, ACCOMODATIONS is the first variable chosen.

Regressions are estimated with OVERALL as the dependent variable and ACCOMODATIONS as an explanatory variable.

Each remaining independent variable is included with ACCOMODATIONS. Hence, 21 regressions are run, each with a different explanatory variable.

The variable with the lowest p-value from these 21 regressions is retained.

In our application, MEALCOURTESY is the second variable chosen.

The procedure continues until no additional variable has a p-value less than a specified cutoff.

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Stepwise Regression Continued



Using .1 as our p-cutoff value, we obtain the following output:.

```
y = Hospital_Satisfaction.OVERALL
X = Hospital_Satisfaction[
    ["ACCOMMODATIONS", "AGE", "GENDER", "MEALCOURTESY", "MEALSATISFACT", "MEALTASTE", "MEALTEMP", "MEALTIMING", "MEALVARIETY", "RECENT",
     "ROOMBEDDING", "ROOMCLEANLINESS", "ROOMDECOR", "ROOMFAMILY", "ROOMNOISE_IN", "ROOMNOISE_OUT", "ROOMPRIVACY", "ROOMRESTROOM", "ROOMSTAFF",
     "ROOMTEMPERATURE", "ROOMTV", "WOM"]]

stepwise_regression(y, X, 0.1, verbose=True)
```

Selected variables:

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Regression Model with Chosen Variables



We can then estimate a regression with the chosen variables.

I named the regression RegStepOut

```
RegStepOut = tprstats.model("cs", "OVERALL~ACCOMMODATIONS+MEALCOURTESY+ROOMBEDDING", Hospital_Satisfaction)
RegStepOut.summary()
```

- It is important to understand that this is not a causal model.
- This was, however, very useful for the Tepper team because it helped then show the hospital manager that there were significant predictors among the variables that were obtained.
- This in turn provided motivation for the manager to undertake surveys of future patients to get a larger sample.

Dep. Variable:	OVERALL	R-squared:	0.381			
Model:	OLS	Adj. R-squared:	0.346			
No. Observations:	56	F-statistic:	10.43			
Covariance Type:	HCl	Prob (F-statistic):	1.77e-05			
	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.6431	0.618	1.040	0.303	-0.598	1.884
ACCOMMODATIONS	0.3129	0.149	2.096	0.041	0.013	0.613
MEALCOURTESY	0.2875	0.102	2.824	0.007	0.083	0.492
ROOMBEDDING	0.2737	0.141	1.937	0.058	-0.010	0.557

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Stepwise Results



Our stepwise regression retains the three variables shown on the previous slide.

Intuitively, why were none of these significant at the 5% or 10% level in the earlier regression when we included all of the variables?

- Correlation among the right-hand-side variables, i.e., multicollinearity.

This model may be a good predictor of hospital satisfaction, but, as noted on the previous slide, we need to be careful not to interpret this as a causal model.

The coefficients have plausible algebraic signs. Good!

But other variables may not have been chosen because they are highly correlated with the variables that were chosen. We now illustrate.

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Calculating Correlations in Python



To confirm that several variables are highly correlated, let's calculate correlations between several pairs of variables.

For the four pairs below, we obtain correlations of .668, .788, .745, and .721.

```
Hospital_Satisfaction['ACCOMMODATIONS'].corr(Hospital_Satisfaction['ROOMRESTROOM'])
np.float64(0.6682818632529341)

Hospital_Satisfaction['ROOMPRIVACY'].corr(Hospital_Satisfaction['ROOMNOISE_IN'])
np.float64(0.7880811517822739)

Hospital_Satisfaction['MEALSATISFACT'].corr(Hospital_Satisfaction['MEALTASTE'])
np.float64(0.7455366864855663)

Hospital_Satisfaction['ROOMCLEANLINESS'].corr(Hospital_Satisfaction['ROOMRESTROOM'])
np.float64(0.731153813885295)
```

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Investigating Variables Not Included by Stepwise Procedure



After estimating a model using the stepwise procedure, we can, if we wish, run a regression including variables not chosen by the stepwise procedure.

Let's estimate the model with the selected variables as well as ROOMRESTROOM.

Dep. Variable:	OVERALL	R-squared:	0.392			
Model:	OLS	Adj. R-squared:	0.344			
No. Observations:	56	F-statistic:	8.626			
Covariance Type:	HC1	Prob (F-statistic):	2.14e-05			
	coef	std err	t	P > t	[0.025	0.975]
Intercept	0.6092	0.595	1.024	0.311	-0.586	1.804
ACCOMODATIONS	0.2354	0.145	1.620	0.111	-0.056	0.527
MEALCOURTESY	0.2641	0.090	2.931	0.005	0.083	0.445
ROOMBEDDING	0.2476	0.157	1.576	0.121	-0.068	0.563
ROOMRESTROOM	0.1382	0.164	0.841	0.405	-0.192	0.468

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Effect of Adding A Variable



- The next slide shows the previous and the new regression.
- Note the change in the coefficient of the accommodations variable and an increase in its p-value when roomrestroom is included.
- Why does addition of roomrestroom have these effects on the coefficient and p-value of accommodations?
- The rating of accommodations and the rating of roomrestroom are correlated. We saw on an earlier slide that the correlation is .668.
- Correlated variables tend to move together.
- When two or more right-hand-side (RHS) variables move together, their individual effects can be difficult to distinguish.
- As previously noted, this is called "Multicollinearity".

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Effect of Adding Correlated Variables [back](#)



Accommodations variable is significant at the 5% level in this regression:

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.6431	0.618	1.040	0.303	-0.598	1.884
ACCOMMODATIONS	0.3129	0.149	2.096	0.041	0.013	0.613
MEALCOURTESY	0.2875	0.102	2.824	0.007	0.083	0.492
ROOMBEDDING	0.2737	0.141	1.937	0.058	-0.010	0.557

When we include Roomrestroom, Accommodations is no longer significant at the 5% or the 10% level. Satisfaction with accommodations is correlated with having a restroom in the room (correlation =.668).

	coef	std err	t	P> t	[0.025	0.975]
Intercept	0.6092	0.595	1.024	0.311	-0.586	1.804
ACCOMMODATIONS	0.2354	0.145	1.620	0.111	-0.056	0.527
MEALCOURTESY	0.2641	0.090	2.931	0.005	0.083	0.445
ROOMBEDDING	0.2476	0.157	1.576	0.121	-0.068	0.563
ROOMRESTROOM	0.1382	0.164	0.841	0.405	-0.192	0.468

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Definition of Multicollinearity



Definition of Multicollinearity: When two or more right-hand-side variables are sufficiently highly correlated with each other that the data do not yield precise estimates of the coefficients of those variables, we have multicollinearity.

A Tepper MBA student (now an alum) suggested a very nice intuitive example of multicollinearity. Her example written in her own words is [included in the appendix to these slides](#).

In analysis of the hospital satisfaction data, we found evidence that some factors measured in the survey appear to affect patient satisfaction. We also found that multicollinearity is present. We would need a substantially bigger survey sample to draw more definitive conclusions about which factors are most important.

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Requiring Variables To Be Included by Stepwise Procedure



In the previous slides, we added a variable to a regression that was not selected by the stepwise procedure. Sometimes we may want to use the stepwise procedure while requiring that one or more variables be included. For example, suppose we want to use the stepwise procedure while requiring that ROOMFAMILY and MEALVARIETY be included.

This can be done using the features option as shown below.

```
y = Hospital_Satisfaction.OVERALL
X = Hospital_Satisfaction[
    ["ACCOMODATIONS", "AGE", "GENDER", "MEALCOURTESY", "MEALSATISFACT", "MEALTASTE", "MEALTEMP", "MEALTIMING",
     "MEALVARIETY", "RECENT", "ROOMBEDDING", "ROOMCLEANLINESS", "ROOMDECOR", "ROOMFAMILY", "ROOMNOISE_IN", "ROOMNOISE_OUT",
     "ROOMPRIVACY", "ROOMRESTROOM", "ROOMSTAFFCOURTESY", "ROOMTEMPERATURE", "ROOMTV", "WOW"]
]

stepwise_regression(y, X, 0.1, features=["ROOMFAMILY", "MEALVARIETY"], verbose=True)
```

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Training Sample and Test Sample



Suppose you have a larger sample than the data set we have. The following is an approach you can use to estimate and evaluate your model.

Suppose you have a larger sample than ours, say 100 observations. Randomly divide your observations into two subsamples. For example, you might select 80% of the observations to use in estimating your model. This is called the training sample.

The remaining 20% of the observations are set aside to test the accuracy of predictions from the estimated model. This is called the test sample.

Then proceed as follows:

- Estimate the model with stepwise regression using the training sample.
- Then, using the explanatory variables from the test sample, predict the values of the dependent variable for the observations in the test sample.
- Compare the predicted values for the test sample to the actual values in the test sample to assess the accuracy of prediction of the estimated model.

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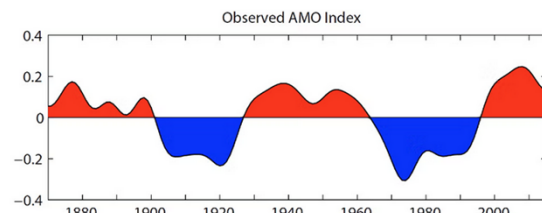
Natural Gas Demand



In a homework assignment, you studied a model of demand for natural gas. That model had sea surface temperature variables.

The graph below shows smoothed data for Atlantic Sea Surface Temperatures.

North Atlantic sea surface temperatures oscillate at multidecadal frequency.



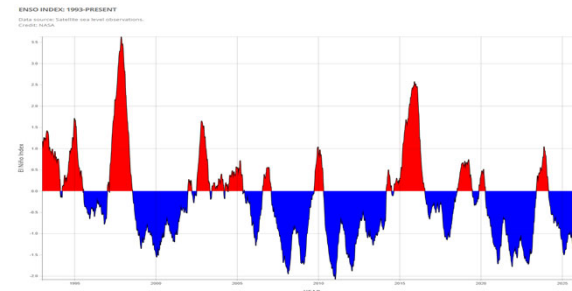
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Equatorial Sea Surface Temperatures



The graph for AMO on the previous slide was for a much longer period: 1870 through 2025. The graph below shows the pattern of variation of ENSO from 1993 to the present. ENSO fluctuates at much higher frequency than AMO and with a much more erratic pattern.



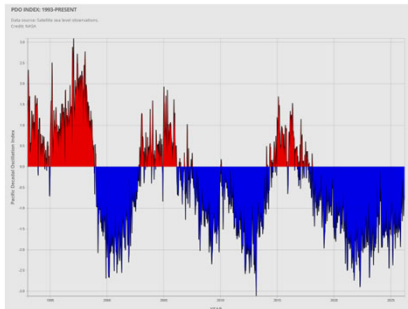
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Pacific Sea Surface Temperatures



Pacific sea surface temperatures oscillate at decadal frequency. The graph below shows Pacific Decadal Oscillation (PDO) from 1993 to the present.



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Definitions of Ocean Surface Temperature Variables



R_AMO is the difference between the maximum and minimum Atlantic sea surface temperature each year.

R_ENSO is the difference between the maximum and minimum Equatorial sea surface temperature each year.

R_AMO_1 is the value of R_AMO in the preceding year. This is called the one-year lag of the variable.

R_AMO_2 is the value of R_AMO two years earlier. This is called the two-year lag of the variable.

R_ENSO_1 and R_ENSO_2 are defined similarly for Equatorial sea surface temperatures.

R_PDO, R_PDO_1, R_PDO_2 denote Pacific Decadal Oscillation variables.

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Illustration of Lagged Variable



The table at the right displays the first few observations of R_AMO, R_AMO_1, and R_AMO_2.

In 1991, R_AMO was .225

Hence, in 1992, R_AMO_1 was .225

In 1993, R_AMO_2 was .225

The table thus shows the R_AMO as well as the one- and two-period lags of AMO.

Be sure you get the idea of a lagged variable. It is straightforward and very important.

Year	R_AMO	R_AMO_1	R_AMO_2
1990	0.445	0.466	0.337
1991	0.225	0.445	0.466
1992	0.288	0.225	0.445
1993	0.155	0.288	0.225
1994	0.289	0.155	0.288
1995	0.426	0.289	0.155

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Stepwise Regression for Natural Gas Model



The command below requires that the logarithms of price and income be included.

```
# Perform stepwise regression to aid in determining which ocean surface temperatures to include.
y = NatGasDat.log_QG

X = NatGasDat[
  [ "log_PG", "log_PCI",
    "R_AMO", "R_AMO_1", "R_AMO_2", "R_ENSO", "R_ENSO_1", "R_ENSO_2"
  ]
]

stepwise_regression(y, X, 0.10, features=["log_PG", "log_PCI"], verbose=True)
```

The stepwise procedure selects which of the following sea surface temperature variables to include: R_AMO, R_AMO_1, R_AMO_2, R_ENSO, R_ENSO_1, R_ENSO_2

We use a p-cutoff= .1.

The above command selects R_AMO, R_AMO_1, and R_ENSO_1:

```
Add R_AMO_1          with p-value 0.025872
Add R_ENSO_1          with p-value 0.066935
Add R_AMO              with p-value 0.044374
```

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Regression Result



The following is the estimated demand function for natural gas including the sea surface temperature variables selected by the stepwise procedure.

Are all coefficients significant at the 5% level?

Interpret the coefficients of log(PG) and Log(PCI).

Do you agree that an increase in any of sea surface temperature variables reduces the demand for natural gas?

Interpret the coefficient of R_AMO.

Hints: 1. Semi-elasticity.

2. Absolute value of the coefficient < .15

OLS Regression Results					
Dep. Variable:	log(QG)	R-squared:	0.876		
Model:	OLS	Adj. R-squared:	0.852		
No. Observations:	32	F-statistic:	64.90		
Covariance Type:	HAC	Prob (F-statistic):	7.48e-14		
	coef	std err	t	P> t	[0.025 0.975]
Intercept	4.4060	0.104	42.252	0.000	4.192 4.620
log(PG)	-0.2538	0.016	-16.149	0.000	-0.286 -0.222
log(PCI)	0.1928	0.028	6.954	0.000	0.136 0.250
R_AMO	-0.1044	0.036	-2.873	0.008	-0.179 -0.030
R_AMO_1	-0.1213	0.027	-4.563	0.000	-0.176 -0.067
R_ENSO_1	-0.0117	0.004	-2.901	0.007	-0.020 -0.003

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Summary of Stepwise Regression



We have seen how to apply stepwise regression.

- Stepwise regression can be a valuable tool in helping you deal with several candidate variables, but it is not a substitute for careful judgment in deciding which variables to include in your regression.
- Stepwise regression is a decision aid, not a decision maker.

We defined multicollinearity and discussed the intuition for why multicollinearity causes imprecise coefficient estimates.

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Summary of Stepwise Regression



We have seen how to apply stepwise regression in two different settings.

In their hospital satisfaction analysis, our students presented to hospital management:

- Their finding that, together, the variables measured in their survey were significant. [This is supported by the p-value = .0255 for their overall regression.]
- The findings of their stepwise regression along with discussion of limitations.
- Their recommendation that a larger survey be undertaken to pinpoint more precisely which factors are most important to overall satisfaction.
- We emphasized that the selected model cannot be considered a causal model.

In the analysis of natural gas demand, we used stepwise regression to aid in choosing sea surface temperature variables while requiring that the logarithms of price and income be included.

- This may well be a causal model because there are well-established scientific reasons for expecting sea surface temperatures to affect climate.
- Also, it is quite plausible that lagged values of the sea surface temperatures have an effect because the changes in sea temperatures may take time to propagate across the continent.

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Multicollinearity Example

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Variables will exhibit multicollinearity when they go hand-in-hand.

Suppose you want to predict how many grams of food a child eats per day. Two variables you might have are age and height. But age and height are themselves correlated during childhood – older kids are taller, and require more nutrition to supply their growing bodies (why teenage boys eat like garbage disposals).

Age and height do have a real relationship with how much a child eats, but each tells only part of the whole story. Included together to explain weight, each may be insignificant by the p-values. Suppose we were to take out *height* and leave *age*. The *age* would be significant and attempting to speak for both variables—as if all the important information were coming from *age* alone, which is not accurate. In reality, we know some very young children are tall for their age, so they eat more.

Thus, multicollinearity causes a problem because it becomes difficult to isolate the effect of individual variables, and difficult to decide which variables to keep and which to remove. This example also illustrates that knowledge of the situation can guide us. In this case, we know both variables are relevant.

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In addition to stepwise regression, other machine learning methods are often used. Examples are

- Lasso Regression
- Ridge Regression
- Elastic Net (which combines the above)

The next two slides describe Lasso and Ridge.

The descriptions below are brief relatively informal.

To set the stage, suppose we have right-hand-side variables $x_1, x_2, \dots, x_k$ where k may be large and where multicollinearity may be present.

Our model is the standard linear equation that we have been studying in multiple regression:

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_k x_k + \varepsilon$$

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Every variable is standardized by subtracting its mean and dividing by its standard deviation.

Lasso regression constrains the sum of the absolute values of the coefficients to be less than some specified value λ_L :

$$|\beta_1| + |\beta_2| + \dots + |\beta_k| \leq \lambda_L$$

The sum of squared errors is minimized subject to the above constraint. As λ_L is reduced, the coefficients shrink, some more than others. (Lasso is called a “shrinkage” estimator.)

If λ_L is reduced far enough some coefficients go to zero (i.e., the variables drop out of the equation).

The approach of using a training sample for estimation followed by evaluation on a test sample is used to investigate the predictive accuracy of the model.

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Ridge regression imposes a constraint on the sum of the squared coefficients:

$$(\beta_1)^2 + (\beta_2)^2 + \dots + (\beta_k)^2 \leq \lambda_R$$

The sum of squared errors is minimized subject to the above constraint. As λ_R is reduced, the coefficients shrink, some more than others.

Strictly speaking Ridge regression is not a variable selection method because it shrinks coefficients, some more than others, but it does not drop variables from the model.

This method requires more computer time to compute than Lasso in samples with a very large number of observations and a very large number of variables.

The approach of using a training sample for estimation followed by evaluation on a test sample is used to investigate the predictive accuracy of the estimated model.

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